Intro to SAR – assignment [35 marks]

Deliverables to include:

[10 marks total]

D1. Screenshot of your 4 dB layers comparing the backscattering values – these will ALL have the 7 colours palette applied and class breaks defined in 5 dB increments (as seen on page 22 of the **Week 2** Tutorial)

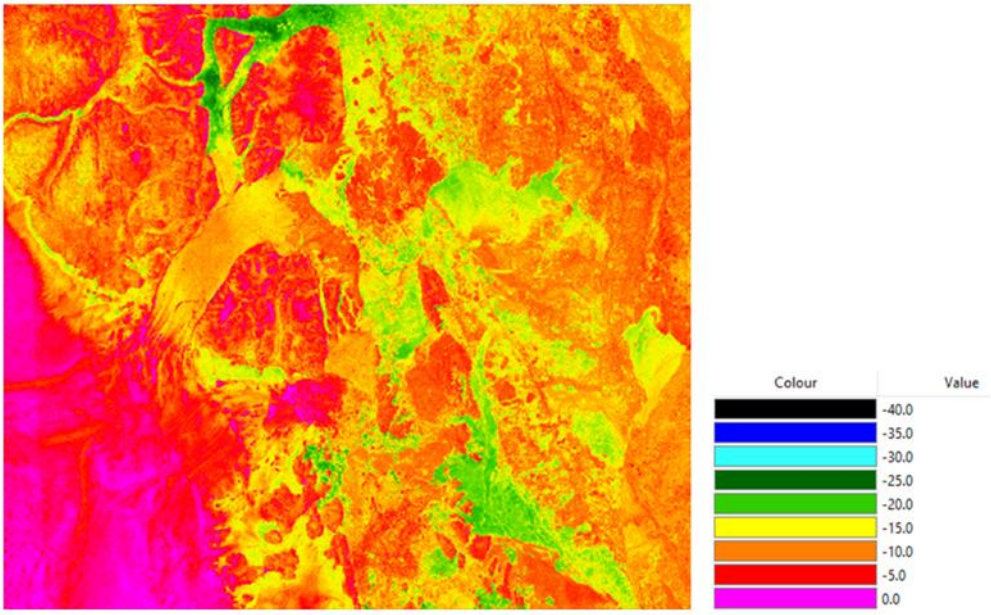

Figure 1. *Backscatter intensity in decibels for the C-band HH polarization, displayed using a 7-colour palette with5 dB intervals from −40 dB to 0 dB.*

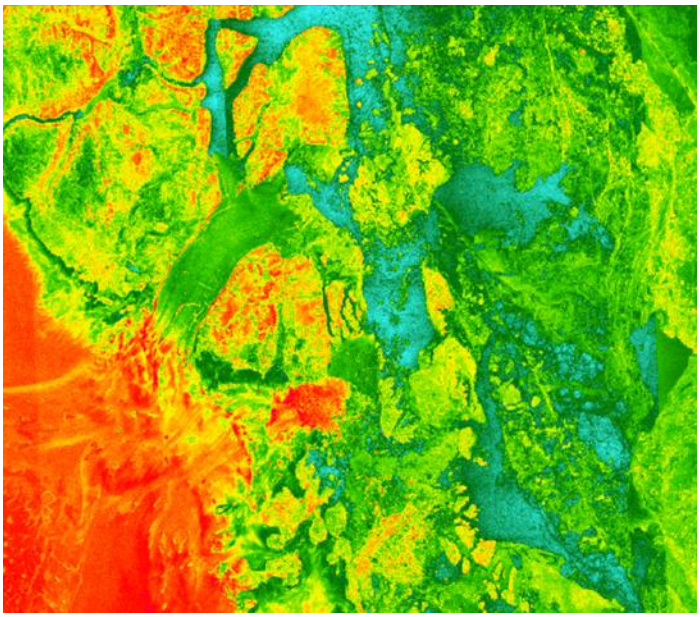

Figure 2. *Backscatter intensity in decibels for the C-band HV polarization, displayed using a 7-colour palette with5 dB intervals from −40 dB to 0 dB.*

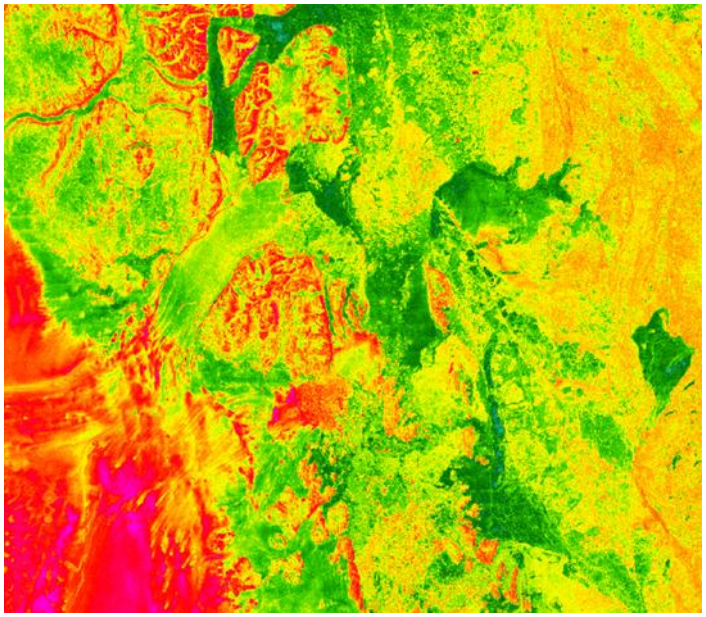

Figure 3. *Backscatter intensity in decibels for the L-band HH polarization, displayed using a 7-colour palette with5 dB intervals from −40 dB to 0 dB.*

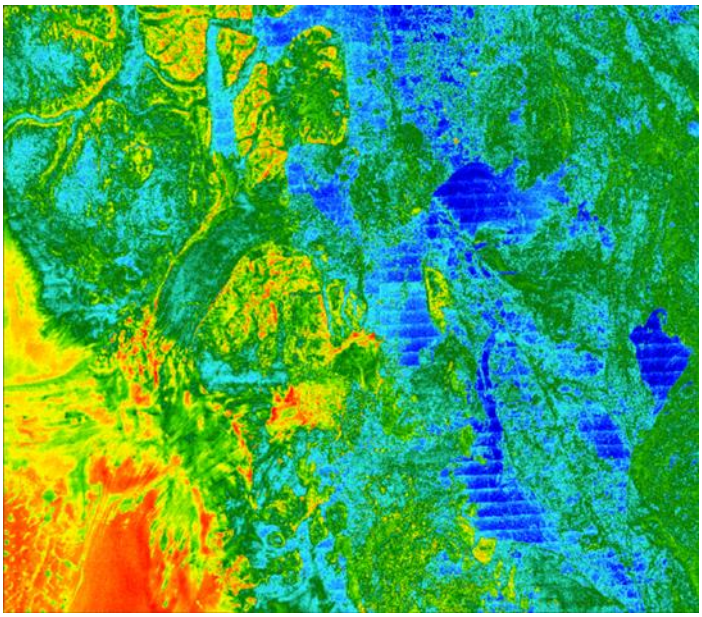

Figure 4. *Backscatter intensity in decibels for the L-band HV polarization, displayed using a 7-colour palette with5 dB intervals from −40 dB to 0 dB.*

D2. Band difference layers (**Week 2**) for:

- C_HH_dB – C_HV_dB
- C_HH_dB – L_HH_dB
  - Apply the same colour palette (7 colours – change the highest class to magenta)
  - stretch to **2σ**
  - distribute the **class breaks evenly**
  - don't forget to include legends – they will be different for these two products. (screenshot the class breaks/colours from the Tables tab)

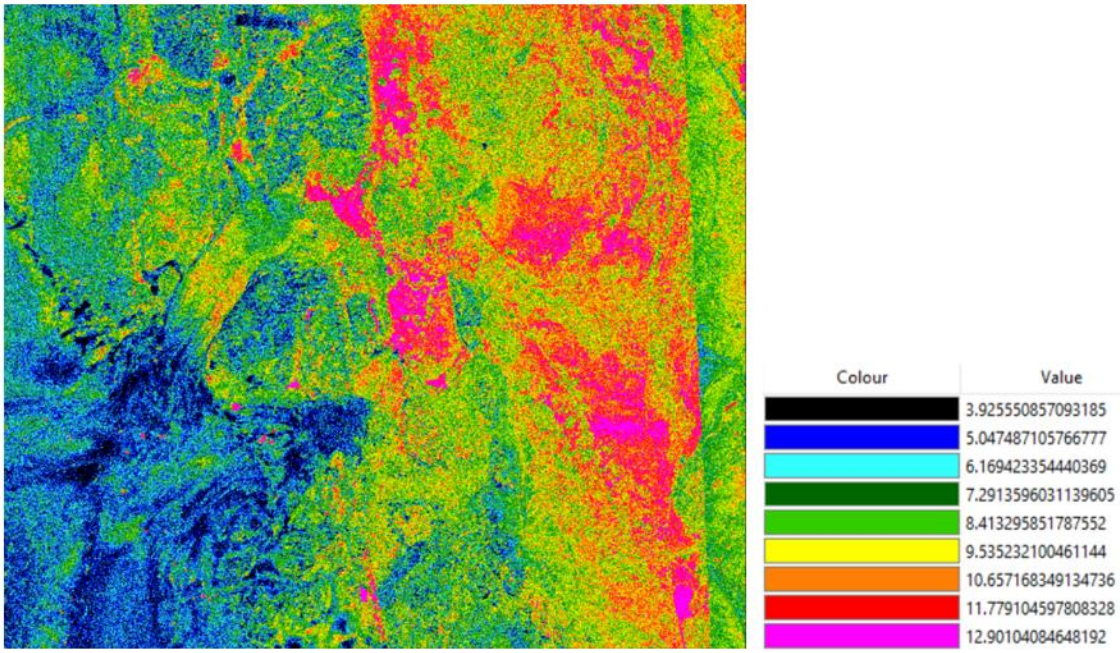

Figure 5. *Difference in backscatter intensity, represented in decibels, between C-band HH and C-band HV polarizations. High values demonstrate a greater difference in the co-polarized and cross-polarized difference, while lower values demonstrate a smaller difference. C-band co-polarization is stronger in all pixels, displaying scattering from rough surfaces, not vegetation.*

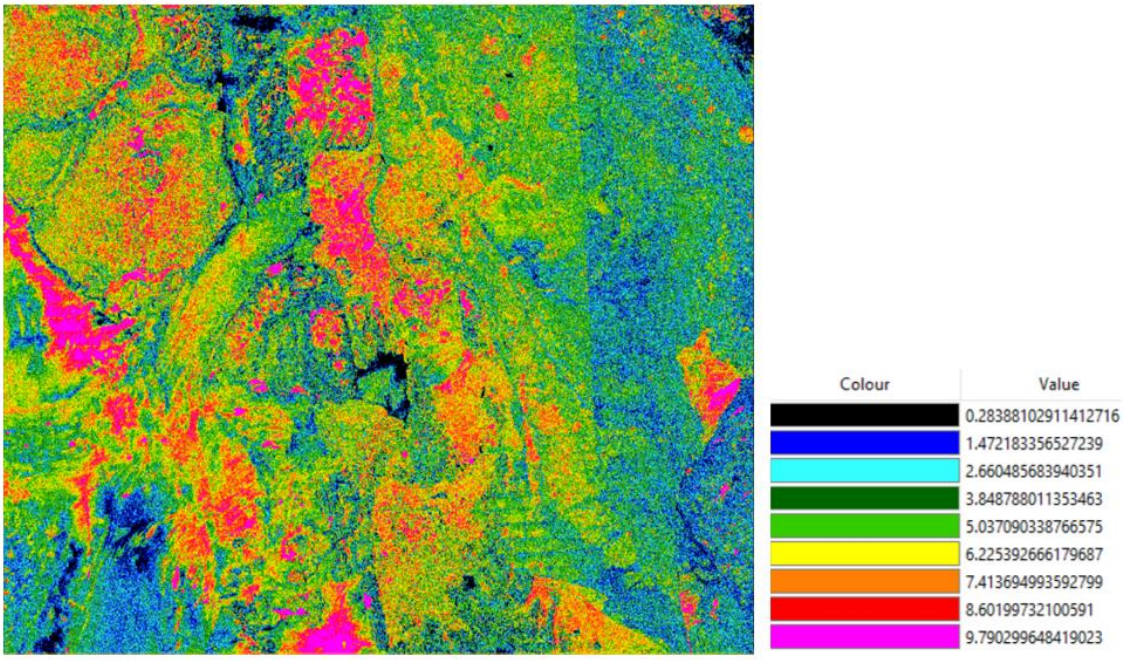

Figure 6. *Difference in backscatter intensity, represented in decibels, between C-band HH and L-band HH polarizations. High values demonstrate a greater difference between the C and L-bands, while lower values demonstrate a smaller difference. The C-band co-polarization is stronger in all pixels, displaying that the surface has little vegetation and a rough surface.*

D3. Additive Backscatter Layer (**Week 3**)

- Apply the same "7 colours" palette
- Stretch to **3 σ**
- Distribute the class breaks evenly
- Change the highest value class to magenta

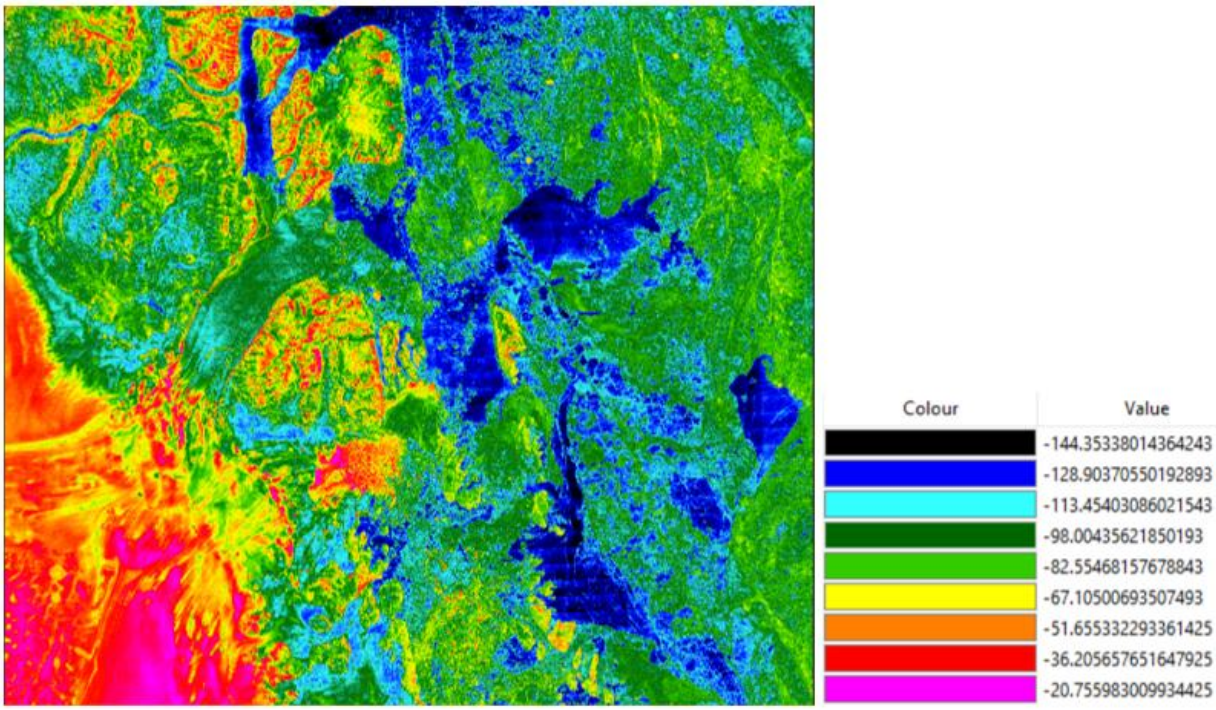

Figure 7. *Additive backscatter layer, with pixel colours demonstrating decibel values. C-band and L-band cross and co-polarization bands were added to distinguish areas with little scattering in all bands.*

D4. Your Google Colab output from **Week 3:**

- Jeffires-Matusita Distance Separability Analysis, including:
  - Number of samples for each class
  - The separability "score" for each pair comparison
- 4 Box and Whiskers plots

Table 1. *Jeffries-Mautista distance separability analysis table demonstrating class names and number of samples.*

Jeffries-Matusita Class Samples	
Class	**Number of Samples**
Sound	699
Icefield	922
Lake	375
Sea Ice 1	299
Sea Ice 2	1755

Table 2. *Jeffries-Mautista distance matrix, comparing the differentiability between classes. Values of 0 represent indistinguishable classes and values of 2 represent unique, separable classes.*

Jeffries-Matusita Distance Matrix					
Classes	**Sound**	**Icefield**	**Lake**	**Sea Ice 1**	**Sea Ice 2**
Sound	---	2.0000	2.0000	1.9093	1.9985
Icefield	2.0000	---	1.9995	2.0000	2.0000
Lake	2.0000	1.9995	---	2.0000	2.0000
Sea Ice 1	1.9093	2.0000	2.0000	---	1.9987
Sea Ice 2	1.9985	2.0000	2.0000	1.9987	---

Table 3. *Jeffries-Mautista distance separability analysis statistics.*

Jeffries-Matusita Summary Statistics		
Statistic	**Value**	**Interpretation**
Mean Distance	1.9906	Mean separability value
Minimum Distance	1.9093	All classes are very distinguishable, in the worst case
Maximum Distance	2.0000	Classes are completely distinguishable, in the best case

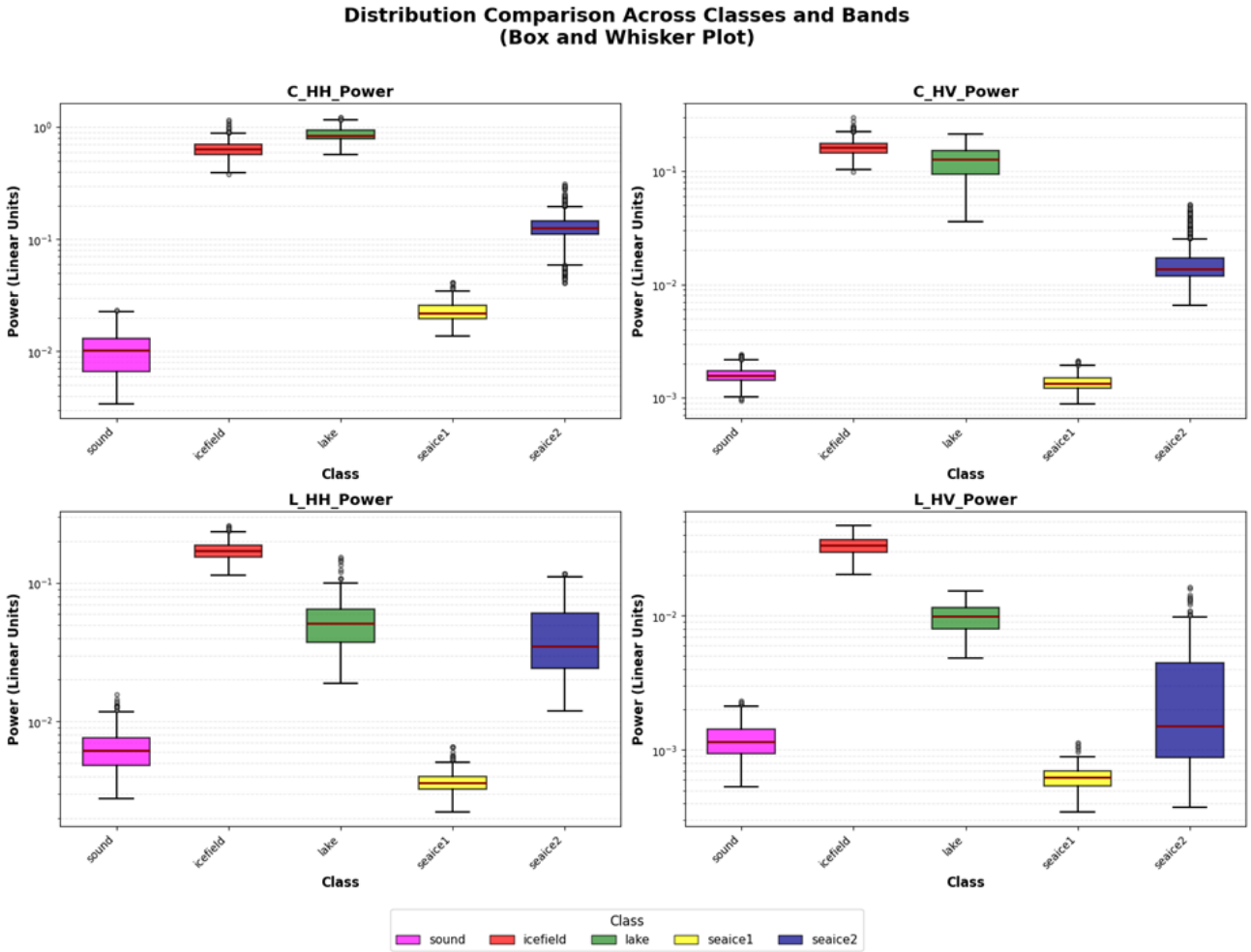

Figure 8. *Box-and-whisker plots of backscatter for classes across SAR bands and cross and co-polarizations. Boxes represent the interquartile range, red lines display medians, whiskers extend 1.5× IQR, and gray dots demonstrate outliers.*

You will refer to these deliverables to help you answer the discussion questions. You can also include any other screenshots of your dB layers (zoomed in to show specific examples) that may support your answers.

Discussion Questions:

[**25 marks total**]

For ALL discussion questions – you can use point form, however you should be using peer reviewed references: in text citations (Evans et al., 2026) as well as listing the full citation at the end of the assignment.

The provided resources are a good place to start (bottom of the Brightspace page)

– you do NOT need to cite the lab materials or lectures for this class.

Q1. What are some important reasons why SAR is preferable to Optical/Multispectral for studies in far north latitudes such as Greenland? Discuss.

[**4 marks – max 250 words**]

In the polar regions, sunlight is limited in the winter months, leading to the choice of SAR over sensors that rely on solar radiation. On average, Greenland is cloudy roughly 80% of the year, causing further complications for optical imaging (Cappelen & Drost Jensen, 2021). Another advantage of SAR is the data given on snow volume and sub-surface properties, due to SAR's cross and co-polarization microwave abilities. While optical remotely sensed data can display information on surface topography, SAR's ability to capture surface roughness, through a synthesized antenna, is an advantage for Northern polar data analysis.

Q2. Regarding Incidence angle: Based on what you see in the Incidence Angle layer, was this pass from Sentinel-1 looking east to west? Or west to east? How do you know?

[**2 marks – very brief**]

For a uniform object, backscatter decreases as the incidence angle increases. Thus, a low incidence angle results in a bright pixel. The image decreases in backscatter, from left to right. From the Sentinel-1 metadata, we can conclude the orbit was moving from South to the North and the sensor was facing East. Therefore, the pass moved from West to East.

Q3. Regarding your individual decibel layer comparisons (refer to **D1** screenshot submitted)

- Which frequency/polarization had the lowest backscatter overall?
- Out of C-band and L-band – which did the best job of differentiating features over the ice field? Discuss why considering backscattering characteristics.

[**3 marks – max 250 words**]

The low frequency L-band with cross-polarization has the lowest overall backscatter. Figure 4 is image with the most pixels designated as dark blue, cyan, and dark green, colours which represent the lower half of the decibel values range. Low backscatter pixels represent smooth areas, such as water, ice sheets, or smooth snowy regions. Long wavelengths are excellent for data regarding snow volume and stratification but lack the ability to provide smaller details, such as minute topography changes.

Shorter wavelengths are better suited to provide information on surface details, due to a shallower penetration depth. Færch et al. (2024) state that "C-band SAR cannot distinguish between rough sea ice and icebergs". The L-band penetrates deeper into the study area's surface but this results in a smoothing effect for contrast between features. Thus, we can

conclude that the C-band is better suited to fine-scale surface detail, such as crevasses or ridges in the ice fields. Pixel values in Figure 2 display a wider range of decibel values, when compared to Figure 1. Thus, the C-band's cross-polarization provides the most information on surface features.

Q4. Regarding your band difference layers:

Out of the 2 band subtraction layers you created (refer to **D2** screenshots submitted):

- Which did the best job of separating features on land/over the icefield? Discuss why considering backscattering characteristics.
- Which did the best job of separating sea ice – both nearshore and off shore. Discuss why considering backscattering characteristics.

[**6 marks – max 500 words**]

The difference layer using the C and L-band horizontal co-polarizations highlights feature separation over land and icefields more effectively than the C-band cross and co-polarization difference image (Figure 5). This image contains a smooth decibel value transitions between features, due to cross-polarizatons sensitivity to volume scattering (Dean & Scharien, 2023), which can be found in snow and ice stratification. Figure 6 exhibits pink, red, and orange pixels, representing high intensity values over the land. Dark coloured pixels are adjacent to bright ones, demonstrating the high surface scattering. Horizontal co-polarization is more sensitive to surface scattering (Dean & Scharien, 2023) and to double bouncing, which allow the sensor to determine sudden, steep elevation changes in the study area. Double bouncing is commonly associated with cities or trees, but in Eastern Greenland, the steep elevation drops include cliffs on land and iceberg edges.

The image of the C-band polarization difference image distinguishes sea ice better than the horizonal co-polarization differences between the C and L-bands. Cross-polarization reveals sub-surface details, particularly important for snow volume and layering. High difference values mean that the HH polarization is much stronger than that of HV, indicating smoother surfaces. Low values represent areas with heterogeneous composition, such as ridges or icebergs. Dean & Scharien (2023) note that cross-polarization in both C and L-bands differentiate between ice sheets and bodies of water, better than their co-polarization counterpart. Stratified land cover experiences far more volume scattering than sea water. Thus, sea ice and sea water are easily classified with cross-polarization. C and L-band scattering do not behave the same way with varying ice thicknesses and types (Mahmud et al., 2022). This information can be used to study near and far offshore sea ice, due to their differing physical properties. Nearshore and land ice

are thinner and smoother, whereas offshore ice tends to be thicker and heterogenous, leading to stronger volume scattering (Mahoney, 2018).

Q5. Regarding your J-M separability analysis (**D4** submitted):

- Which (if any) of the class pairs has poor separability?
- Do your boxplots agree with that analysis?
- Discuss why these classes may be hard to separate considering probable backscattering characteristics.

[**4 marks – max 250 words**]

No classes have poor separability. The minimum Jeffries-Matusita distance value is 1.9093, where a value of 2.0 between two classes represent unique, completely separable classes (see Table 2). In the worst-case scenario, for this study, two classes are very distinguishable. The mean J-M distance value is 1.9906 and the maximum is 2.0000 (Table 3). Therefore, it is clear all classes are highly unique and separable.

The red median lines for all five classes are clearly distinct, further confirming the high separability between classes.

It is known, from the discussion above, how polarization behaves differently with various surface types. It can be inferred that salt and fresh water bodies must scatter differently but as they are both water bodies, they may exhibit similar intensity characteristics. The same reasoning can apply to various sea ice chunks. These similar characteristics can cause misclassification of class types.

Q6. Regarding your boxplots (**D4** submitted):

- Seaice1 and Seaice 2 look very different across both bands/frequencies – discuss some reasons why considering backscattering and location hint – think of surface roughness)
    - [**3 marks – max 200 words**]
- Which band(s)/frequencies(s) – if any – are able to separate the supraglacial lakes from the icefield? Discuss why the lake might have lower backscatter at L-band compared to C-band (hint – think about ability to penetrate)
    - [**3 marks – max 200 words**]

Sea ice 2 consistently has higher backscatter intensity values than sea ice 1. Though, in the C-band polarizations, the gap between the two sea ice classes is smaller (see Figure 8). The sea ice 1 class represents ice chunks that reside between land, in a narrow channel, while sea ice 2 is taken from offshore ice. With the same reasoning from previous questions, we can conclude that smooth, nearshore ice will behave differently than rough, offshore ice. These behaviours change as we move across polarizations and bands.

The L-band polarizations discern the lake class from the icefield better than those of the C-bands. The L-bands longer wavelength allows for a deeper subsurface penetration. As stated by Dean & Scharien (2012), cross-polarization is better suited for supraglacial lakes in winter in both the L and C-bands. Though, a longer wavelength, paired with cross-polarization, is the most optimal choice for lake analysis in the North.

References

Cappelen, J., & Drost Jensen, C. (2021). Climatological standard normals 1991–2020 – Greenland (DMI Report 21-12). Danish Meteorological Institute. Retrieved from https://www.dmi.dk/fileadmin/Rapporter/2021/DMI_report_21_12_Greenland.pdf (dmi.dk)

Dean, C., & Scharien, R. K. (2023, December 13). *Comparison of C- and L-band synthetic aperture radar for enhanced seasonal supraglacial lake detection in Northeast Greenland* [Poster presentation]. American Geophysical Union Fall Meeting 2023, San Francisco, CA, United States.

Færch, L., Høyer, J. L., & Tonboe, R. T. (2024). *Mapping icebergs in sea ice: An analysis of seasonal SAR backscatter at C- and L-band*. *Remote Sensing of Environment, 309,* 114868. https://doi.org/10.1016/j.rse.2024.114868Ddcdc

Mahoney, A. R. (2018). *Landfast sea ice in a changing Arctic*. NOAA Arctic Report Card. https://arctic.noaa.gov/report-card/report-card-2018/landfast-sea-ice-in-a-changing-arctic/

Mahmud, M. S., Nandan, V., Singha, S., Howell, S. E. L., Geldsetzer, T., Yackel, J., & Montpetit, B. (2022). C- and L-band SAR signatures of Arctic sea ice during freeze-up. *Remote Sensing of Environment, 276,* 113129. https://doi.org/10.1016/j.rse.2022.113129